5.3.18 Temperature sensor characteristics

Table 56. Temperature sensor characteristics

Symbol	Parameter	Min	Typ	Max	Unit
T_L[(1)]	V_{SENSE} linearity with temperature (from V_{sensor} voltage)	-	-	3	°C
Avg_Slope[(2)]	Average slope (from V_{sense} voltage)	-	2.14	-	mV/°C
V_{30}[(3)]	Voltage at 30° C (± 1 ° C)	-	0.65	-	V
t_{start_run}[(1)]	Startup time in Run mode (buffer startup)	-	-	25.2	µs
t_{S_temp}[(1)]	ADC sampling time when reading the temperature	13	-	-	µs
I_{sens}[(1)]	Sensor consumption	-	0.18	0.29	µA
$I_{sensbuf}$[(1)]	Sensor buffer consumption	-	3.8	6.5	µA

1. *Specified by design - Not tested in production.*
2. *Evaluated by characterization - Not tested in production.*
3. *Measured at V_{DDA} = 3.3 V ± 10 mV. The V_{30} ADC conversion result is stored in the TS_CAL1.*

Table 57. Temperature sensor calibration values

Symbol	Parameter	Memory address
TS_CAL1	Temperature sensor raw data acquired value at 30 °C, V_{DDA} = 3.3 V	0x08FF F814 - 0x08FF F815
TS_CAL2	Temperature sensor raw data acquired value at 140 °C, V_{DDA} = 3.3 V	0x08FF F818 - 0x08FF F819

5.3.19 Digital-to-analog converter characteristics (DAC)

Table 58. DAC characteristics

Specified by design and not tested in production.

Symbol	Parameter	Conditions		Min	Typ	Max	Unit
V_{DDA}	Analog supply voltage	-		2.7	-		V
V_{REF+}	Positive reference voltage	-		2.5	-	V_{DDA}	V
V_{REF-}	Negative reference voltage	-		-	V_{SSA}	-	V
R_L	Resistive Load	DAC output buffer ON	connected to V_{SSA}	5	-	-	KΩ
			connected to V_{DDA}	25	-	-	KΩ
R_O	Output Impedance	DAC output buffer OFF		10.3	13.00	16	KΩ
R_{BON}	Output impedance sample and hold mode, output buffer ON	DAC output buffer ON	V_{DD} = 2.7 V	-	-	1.6	KΩ
R_{BOFF}	Output impedance sample and hold mode, output buffer OFF	DAC output buffer OFF	V_{DD} = 2.7 V	-	-	17.8	KΩ
C_L	Capacitive Load	DAC output buffer OFF		-	-	50	pF
C_{SH}	Capacitive Load	Sample and Hold mode			0.10	1	µF
V_{DAC_OUT}	Voltage on DAC_OUT output	DAC output buffer ON		0.2	-	V_{DDA} −0.2	V
		DAC output buffer OFF		0	-	V_{REF+}	V